STPM 2024 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $g = 9.81$ m s^{-2}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. Two point charges, $+Q$ and $-Q$, are placed at points X and Y respectively. Which is true about the electric field and electric potential at the midpoint between points X and Y?

	Electric field	*Electric potential*
A	Towards point Y	Non-zero
B	Towards point X	Zero
C	Towards point Y	Zero
D	Towards point X	Non-zero

Answer: C

The field of $+Q$ points away from X and the field of $-Q$ points towards Y: both point towards Y.

$$V = \frac{kQ}{d} + \frac{k(-Q)}{d} = 0$$

Explanation: Field is a vector (adds); potential is a scalar (cancels here).

2. A drop of oil of mass 1.63×10^{-15} kg is suspended stationary between two horizontal metal plates that are separated by 1.0 cm. If the electric potential of the upper plate is +100 V and that of the lower plate is −100 V, what is the magnitude of charge of the oil?

A 1.6×10^{-16} C B 8.0×10^{-17} C C 1.6×10^{-18} C D 8.0×10^{-19} C

Answer: D

$$E = \frac{V}{d} = \frac{100 - (-100)}{0.010} = 2.0 \times 10^4 \text{ V m}^{-1}$$

$$qE = mg \Longrightarrow q = \frac{(1.63 \times 10^{-15})(9.81)}{2.0 \times 10^4} = 8.0 \times 10^{-19} \text{ C}$$

Explanation: The p.d. between the plates is 200 V, not 100 V.

3. Which is true about the effect of inserting a dielectric in a charged parallel plate capacitor?

A The capacitance increases
B The charge stored increases
C The electric field between the plates increases
D The potential difference across the plates increases

Answer: A

$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$: inserting a dielectric ($\varepsilon_r > 1$) increases the capacitance

Explanation: If the capacitor is isolated, Q is constant while E and V decrease.

4. A wire of diameter 3.00×10^{-3} m has a current of 10.0 A. If the drift speed is 4.00×10^{-5} m s^{-1}, what is the density of free electrons in the wire?

A 1.66×10^{26} electrons m^{-3}
B 6.62×10^{26} electrons m^{-3}
C 5.52×10^{28} electrons m^{-3}
D 2.21×10^{29} electrons m^{-3}

Answer: D

$$A = \pi(1.50 \times 10^{-3})^2 = 7.07 \times 10^{-6} \text{ m}^2$$

$$n = \frac{I}{Av_d e} = \frac{10.0}{(7.07 \times 10^{-6})(4.00 \times 10^{-5})(1.60 \times 10^{-19})} = 2.21 \times 10^{29} \text{ m}^{-3}$$

Explanation: Use the radius, half the diameter.

5. A battery of e.m.f., ε, and internal resistance, r, is connected in series with a bulb of resistance R. What is the power supplied by the battery?

A $\frac{\varepsilon^2 R}{(R+r)^2}$ B $\frac{\varepsilon^2}{R+r}$ C $\frac{\varepsilon^2 r}{R^2}$ D $\frac{\varepsilon^2 R}{r^2}$

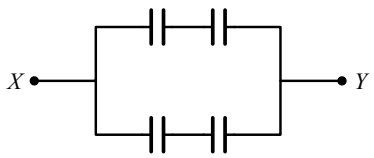
Answer: B

$$I = \frac{\varepsilon}{R+r}; \text{ power supplied} = \varepsilon I = \frac{\varepsilon^2}{R+r}$$

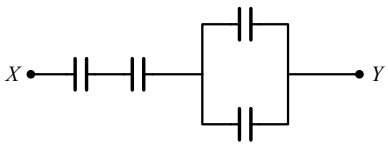
Explanation: Option A is the power delivered to the bulb only, I^2R.

6. Four identical capacitors are connected in four different arrangements. Which arrangement produces the highest value of capacitance between points X and Y?

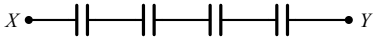
A



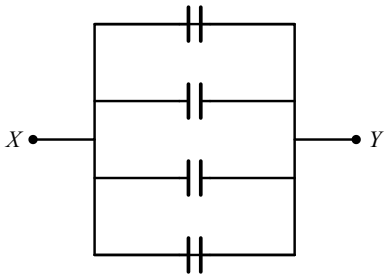
B



C



D

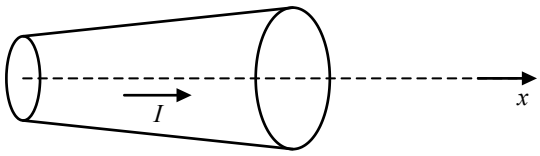


Answer: D

A: $2 \times \frac{C}{2} = C$; B: $\frac{C}{2}$ in series with $2C = 0.4C$; C: $\frac{C}{4}$; D: all four in parallel $= 4C$.

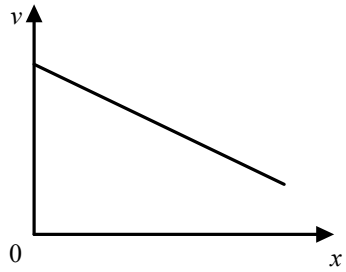
Explanation: Capacitors in parallel add directly.

7. The diameter of a metal conductor uniformly increases along the distance, x, as shown in the diagram below.

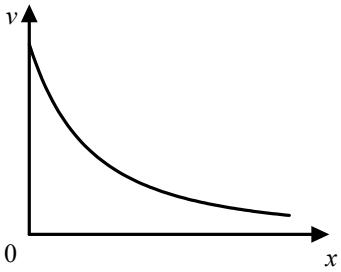


If a constant current, I, flows along the conductor, which graph shows the variation of the drift velocity, v, of the free electrons with x?

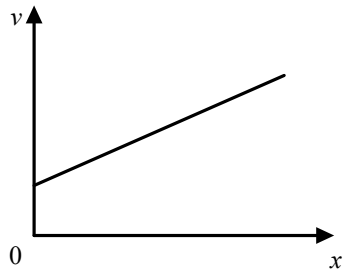
A



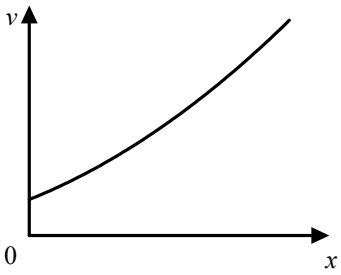
B



C



D



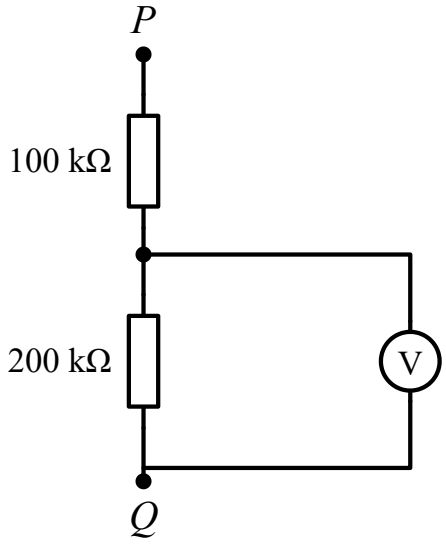
Answer: B

$$v = \frac{I}{nAe} \propto \frac{1}{A} \propto \frac{1}{d^2}$$

The diameter increases linearly with x, so v falls non-linearly, steeply at first (B).

Explanation: $v \propto 1/(a + bx)^2$: a decreasing curve, not a straight line.

8. The arrangement of two resistors with a voltmeter of resistance 30.0 kΩ is shown in the diagram below.



If the potential difference across P and Q is 300 V, what is the reading of the voltmeter?

A 26.1 V B 30.0 V C 62.1 V D 69.2 V

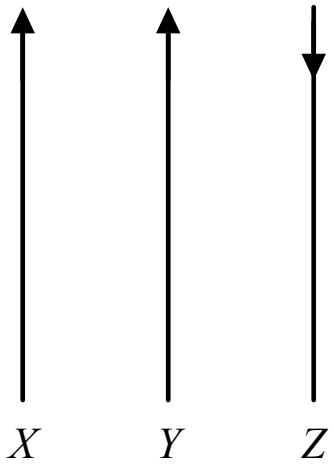
Answer: C

$$200\ \text{k}\Omega \parallel 30.0\ \text{k}\Omega = \frac{(200)(30.0)}{230} = 26.1\ \text{k}\Omega$$

$$V = 300 \times \frac{26.1}{100 + 26.1} = 62.1\ \text{V}$$

Explanation: Without the voltmeter the reading would be 200 V; the voltmeter loads the circuit.

9. Three parallel current-carrying conductors, X, Y and Z, of equal current, are arranged as shown in the diagram below.



If the distance between each conductors are equal, which statement is true about the movement of the conductors?

A X moves to the left away from Y.
B Y moves to the right away from X.
C Z moves to the right away from Y.
D All conductors are not moving.

Answer: C

Like currents attract, unlike currents repel. Z (current down) is repelled by Y and by X, so it moves to the right away from Y (C).

Explanation: Y is pulled left by X and pushed left by Z; X is pulled right by Y more strongly than Z pushes it left.

10. In Thomson's experiment, a beam of electrons travels at 6.8×10^7 m s^{-1} and bends into a circular path of radius 4.0 cm in a magnetic field of 3.0×10^{-2} Wb m^{-2} which perpendicular to its path. What is the value of $\frac{e}{m}$?

A 5.7×10^{10} C kg^{-1}
B 9.1×10^7 C kg^{-1}
C 5.1×10^7 C kg^{-1}
D 8.2×10^4 C kg^{-1}

Answer: A

$$evB = \frac{mv^2}{r} \Longrightarrow \frac{e}{m} = \frac{v}{Br} = \frac{6.8 \times 10^7}{(3.0 \times 10^{-2})(0.040)} = 5.7 \times 10^{10} \text{ C kg}^{-1}$$

Explanation: 1 Wb m^{-2} = 1 T.

11. The Hall voltage in a semiconductor plate does **not** depend on

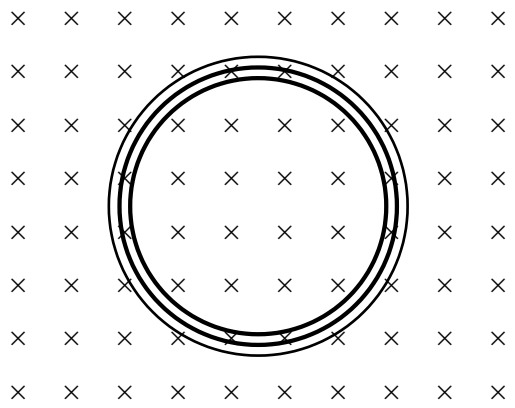
A the length of the semiconductor plate
B the current density of the semiconductor plate
C the magnetic field applied to the semiconductor plate
D the density of the charge carriers in the semiconductor plate

Answer: A

$V_H = \frac{BI}{nqt}$: it depends on B, the current (density) and n, but not on the length of the plate (A)

Explanation: t is the thickness along **B**.

12. A circular coil of 50 turns and radius 5.0 cm is placed in a uniform magnetic field of magnetic flux density 4.00 T as shown in the diagram below.



If the resistance of the coil is 2.0 Ω, which is the magnitude and direction of the induced current in the coil when the magnetic flux density decreases to zero in 0.50 s?

	Magnitude	*Direction*
A	1.57 A	Anticlockwise
B	3.14 A	Anticlockwise
C	1.57 A	Clockwise
D	3.14 A	Clockwise

Answer: C

$$|\varepsilon| = NA\frac{\Delta B}{\Delta t} = (50)\pi(0.050)^2\frac{4.00}{0.50} = 3.14 \text{ V}$$

$$I = \frac{3.14}{2.0} = 1.57 \text{ A}$$

The flux into the page decreases, so the induced current makes a field into the page: clockwise.

Explanation: Lenz's law: the induced current opposes the change in flux.

13. A solenoid, S_2, is inserted into another solenoid, S_1. The current in S_1 is immediately reduced from 15 A to zero in 0.30 s. If an e.m.f. induced in S_2 is 2.5 V, what is the mutual inductance between S_1 and S_2?

A 0.050 H B 0.56 H C 1.8 H D 11 H

Answer: A

$$M = \frac{\varepsilon}{\mathrm{d}I/\mathrm{d}t} = \frac{2.5}{15/0.30} = 0.050 \text{ H}$$

Explanation: The rate of change of current is 50 A s^{-1}.

14. An alternating power supply, $V = 20 \sin 4\pi t$, is connected to a resistor of resistance 9.0 Ω. Which statement is true about the circuit?

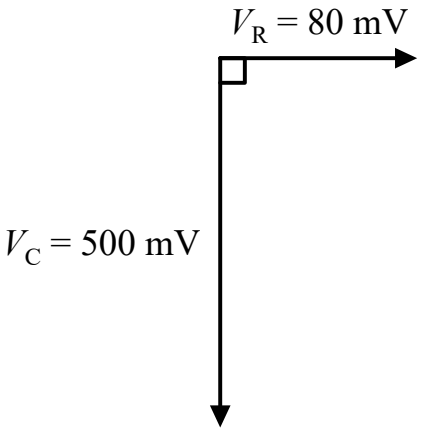
A The root mean square voltage is 28 V.
B The instantaneous power at 0.50 s is zero.
C The mean power dissipated in the resistor is 44 W.
D The voltage is minimum when the current is maximum.

Answer: B

At $t = 0.50$ s, $V = 20 \sin 2\pi = 0$, so the instantaneous power $V^2/R = 0$ (B)

Explanation: $V_{rms} = 20/\sqrt{2} = 14$ V; mean power = $\frac{14.1^2}{9.0}$ = 22 W; in a resistor V and I are in phase.

15. A phasor diagram of V_R and V_C in RC circuit is shown in the diagram below.



If the current flow is 2.0 mA, what is the impedance of the circuit?

A 40.0 Ω B 210 Ω C 247 Ω D 253 Ω

Answer: D

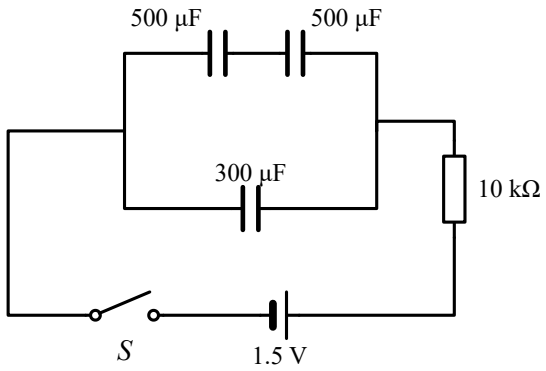
$$V = \sqrt{(80^2 + 500^2)} = 506 \text{ mV}$$

$$Z = \frac{V}{I} = \frac{506 \times 10^{-3}}{2.0 \times 10^{-3}} = 253 \ \Omega$$

Explanation: Add the phasors as vectors, not arithmetically.

Section B [15 marks]

16. Three uncharged capacitors, a resistor, a cell and a switch, S, are connected as shown in the circuit diagram below.



(a) Calculate the equivalent capacitance in the circuit. [3]

Answer

Two 500 µF in series: $\frac{(500)(500)}{500 + 500} = 250$ µF — 1

In parallel with 300 µF: $C = 250 + 300$ — 1

$C = 550$ µF — 1

3

Explanation: Series first, then parallel.

(b) When S is closed, determine the potential difference across the resistor at 10 s. [4]

Answer

$\tau = RC = (10 \times 10^3)(550 \times 10^{-6}) = 5.5$ s — 1

$V_R = V_0 e^{-t/RC}$ — 1

$V_R = 1.5\ e^{-10/5.5}$ — 1

$V_R = 0.243$ V — 1

4

Explanation: Equivalently, $V_C = 1.5(1 - e^{-10/5.5}) = 1.257$ V and $V_R = 1.5 - 1.257$.

(c) If all the capacitors are fully charged, state the potential difference across the resistor. [1]

Answer

Zero (no current flows) — 1

1

Total for question **8**

Explanation: The whole 1.5 V is then across the capacitors.

17. A light bulb has a power rating of 100 W, 240 V. If the resistance per unit length is 280 Ω m^{-1} and the radius of the filament wire is 23 μm, calculate

(a) the length of the filament [2]

Answer

$R = \frac{V^2}{P} = \frac{240^2}{100} = 576\ \Omega$; $l = \frac{576}{280}$ 1

$l = 2.06$ m 1

2

Explanation: Resistance per unit length × length = resistance.

(b) the current that flows through the filament [2]

Answer

$I = \frac{P}{V} = \frac{100}{240}$ 1

$I = 0.417$ A 1

2

Explanation: Assume the bulb works at its rated voltage.

(c) the resistivity of the filament [3]

Answer

$\rho = \frac{RA}{l} = (\frac{R}{l})A$ 1

$\rho = (280)\pi(23 \times 10^{-6})^2$ 1

$\rho = 4.65 \times 10^{-7}\ \Omega$ m 1

3

Total for question **7**

Explanation: The length cancels: only the resistance per unit length is needed.

Section C [30 marks]

18. (a)

(i) Define *electric field strength* of an isolated point charge. [1]

Answer

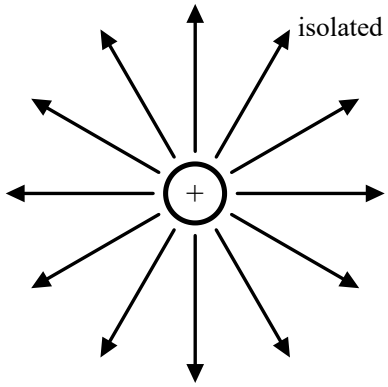
The force per unit positive charge placed at that point in the field of the point charge, $E = \frac{F}{q}$ 1

1

Explanation: For a point charge $E = \frac{Q}{4\pi\varepsilon_0 r^2}$.

(ii) Sketch the electric field pattern for an isolated point charge. [2]

Answer



Radial lines starting on the charge 1

Arrows outwards for a positive charge (inwards for a negative charge) 1

2

Explanation: Lines are closer together near the charge, where the field is stronger.

(iii) Describe the motion of a point charge in a uniform electric field. [3]

Answer

The charge experiences a constant force qE, so it has a constant acceleration 1

If it starts at rest or moves along the field, it moves in a straight line with uniform acceleration (positive charge along **E**, negative charge opposite) 1

If it enters perpendicular to the field, it follows a parabolic path 1

3

Explanation: The motion is like a projectile under gravity.

(b) A spherical shell conductor of radius 15 cm consists of charges +2.0 μC and −5.0 μC.

(i) Determine the electric field strength at 5.0 cm away from the conductor surface and its direction. [6]

Answer

Net charge $Q = +2.0 - 5.0 = -3.0$ μC — 1

$r = 15 + 5.0 = 20$ cm $= 0.20$ m from the centre — 1

$E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$ — 1

$E = \dfrac{(9.0 \times 10^9)(3.0 \times 10^{-6})}{(0.20)^2}$ — 1

$E = 6.75 \times 10^5$ N C^{-1} — 1

directed radially towards the centre of the sphere (net charge negative) — 1

6

Explanation: Outside a charged sphere, the charge behaves as if concentrated at the centre.

(ii) If another spherical shell conductor of radius 15 cm and charge +5.0 μC is brought 5.0 cm to the initial spherical shell conductor, determine the electric potential on the surface of the conductor. [3]

Answer

Potential due to its own charge on the surface of the +5.0 μC shell: $\dfrac{(9.0 \times 10^9)(5.0 \times 10^{-6})}{0.15} = 3.00 \times 10^5$ V — 1

Potential there due to the −3.0 μC shell (0.20 m from its centre): $\dfrac{(9.0 \times 10^9)(-3.0 \times 10^{-6})}{0.20} = -1.35 \times 10^5$ V — 1

$V = 3.00 \times 10^5 - 1.35 \times 10^5 = 1.65 \times 10^5$ V — 1

3

Total for question **15**

Explanation: Taken at the point of the second shell's surface nearest the first shell, treating each shell as a point charge at its centre. If the question means the first shell's nearest surface point instead: $-1.80 \times 10^5 + 2.25 \times 10^5 = 4.5 \times 10^4$ V.

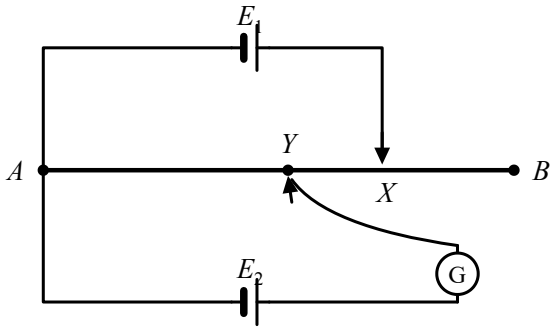
19. (a) Explain the effect of the internal resistance of a battery in a circuit. [4]

Answer

When a current flows, part of the e.m.f. is used to drive charge through the internal resistance (lost volts Ir)	1
The terminal p.d. $V = \varepsilon - Ir$ is less than the e.m.f.	1
The terminal p.d. falls as the current increases (as the external resistance decreases)	1
Energy is dissipated as heat inside the battery, so less power is delivered to the external circuit; it also limits the maximum current	1
	4

Explanation: Maximum power is delivered when the external resistance equals r.

(b) Two batteries, E_1 and E_2, and a galvanometer, G, are connected to a sliding wire AB of 200 cm length with the resistance of 20.0 Ω as shown in the circuit diagram below.



E_2 is 2.00 V. When the distances of AX is 150 cm and AY is 100 cm, there is no deflection of the galvanometer.

(i) Calculate the potential difference across AX. [2]

Answer

Potential gradient $= \frac{2.00}{100} = 0.0200 \text{ V cm}^{-1}$	1
$V_{AX} = (0.0200)(150) = 3.00 \text{ V}$	1
	2

Explanation: At balance the p.d. across AY equals the e.m.f. of E_2.

(ii) If point X is moved to point B, the new distance of AY is 125 cm, calculate the potential difference across AB. [2]

Answer

New gradient $= \frac{2.00}{125} = 0.0160 \text{ V cm}^{-1}$ — 1

$V_{AB} = (0.0160)(200) = 3.20 \text{ V}$ — 1

2

Explanation: A longer external wire draws less current, so the gradient falls.

(iii) Calculate the internal resistance of E_1. [4]

Answer

$R_{AX} = \frac{150}{200} \times 20.0 = 15.0\ \Omega$, $I_1 = \frac{3.00}{15.0} = 0.200 \text{ A}$ — 1

$R_{AB} = 20.0\ \Omega$, $I_2 = \frac{3.20}{20.0} = 0.160 \text{ A}$ — 1

$E_1 = 3.00 + 0.200r = 3.20 + 0.160r$ — 1

$r = 5.0\ \Omega$ (and $E_1 = 4.0 \text{ V}$) — 1

4

Explanation: Two readings of terminal p.d. and current give both ε and r.

(c)

(i) State **two** differences between an ohmic conductor and non-ohmic conductor. [2]

Answer

An ohmic conductor obeys Ohm's law: $I \propto V$ (straight-line I–V graph through the origin); a non-ohmic conductor does not — 1

An ohmic conductor has constant resistance (at constant temperature); the resistance of a non-ohmic conductor changes with the p.d./current — 1

2

Explanation: A non-ohmic conductor may also conduct differently in the two directions.

(ii) Give an example for non-ohmic conductor. [1]

Answer

A semiconductor diode (or a filament lamp, a thermistor) — 1

1

Total for question **15**

Explanation: A diode conducts mainly in one direction.

20. (a) Define *magnetic flux density* experienced by a moving charge. [1]

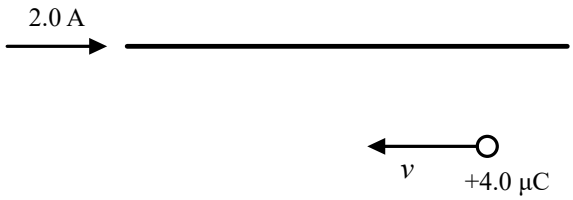
Answer

The force per unit charge per unit velocity on a charge moving perpendicular to the field, $B = \frac{F}{qv \sin \theta}$ 1

1

Explanation: Unit: tesla (T).

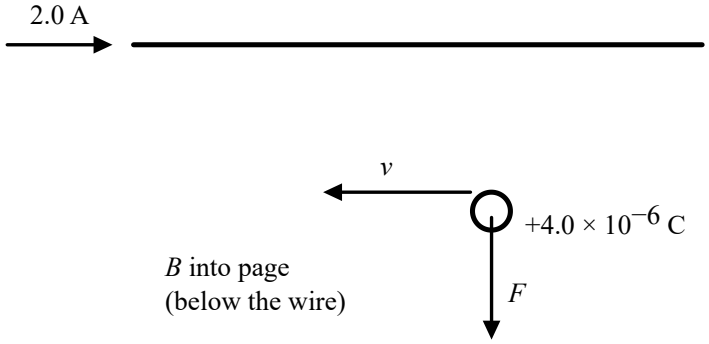
(b) A charge of +4.0 μC is moving at a constant velocity, v, in the direction parallel to a long straight current-carrying wire of 2.0 A as shown in the diagram below.



The distance between the wire and the charge is 1.0 cm.

(i) Sketch a diagram to show the direction of the force on the charge. [2]

Answer



Magnetic field at the charge into the page (right-hand grip rule) 1

Force on the charge directed away from the wire (downwards) 1

2

Explanation: Equivalently, a positive charge moving opposite to the current behaves like an antiparallel current, so it is repelled.

(ii) If the velocity of the charge is 2.18×10^6 m s^{-1}, determine the force on the charge. [4]

Answer

$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7})(2.0)}{2\pi(0.010)}$ 1

$B = 4.0 \times 10^{-5}$ T 1

$F = qvB = (4.0 \times 10^{-6})(2.18 \times 10^6)(4.0 \times 10^{-5})$ 1

$F = 3.49 \times 10^{-4}$ N 1

4

Explanation: **v** is perpendicular to **B**, so sin $\theta = 1$.

(iii) If the magnetic field strength is uniform, describe the motion of the charge. [2]

Answer

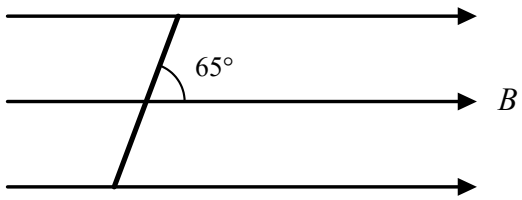
The force is always perpendicular to the velocity, so the speed stays constant 1

The charge moves in a circle (uniform circular motion) of radius $r = \frac{mv}{qB}$ 1

2

Explanation: The magnetic force does no work.

(c) A circular coil of 20 turns with a diameter of 5.0 cm is placed in a uniform magnetic field of 5.0 mT at an angle of 65° from the magnetic field as shown in the diagram below.



(i) Determine the magnetic flux through the coil. [4]

Answer

$A = \pi(0.025)^2 = 1.96 \times 10^{-3}$ m^2 1

Angle between **B** and the normal to the coil = 90° − 65° = 25° 1

$\Phi = BA \cos 25° = (5.0 \times 10^{-3})(1.96 \times 10^{-3}) \cos 25°$ 1

$\Phi = 8.9 \times 10^{-6}$ Wb (flux linkage $N\Phi = 1.78 \times 10^{-4}$ Wb) 1

4

Explanation: The 65° is measured between the plane of the coil and **B**.

(ii) If the coil rotates about an axis in the plane of the coil, state **two** effects experienced by the coil. [2]

Answer

The flux linkage changes continuously, so an alternating e.m.f. (and current, if the circuit is closed) is induced in the coil	1
The induced current in the field produces a torque opposing the rotation (Lenz's law), so work must be done to keep it rotating (the coil also heats up)	1
	2
Total for question	**15**

Explanation: This is the principle of the a.c. generator.